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Controlling Indoor Acoustic Pollution by Optimal Placement of Absorbing Material

Helmholtz modelling, adjoint-based optimisation and a fractal wall for a concert hall

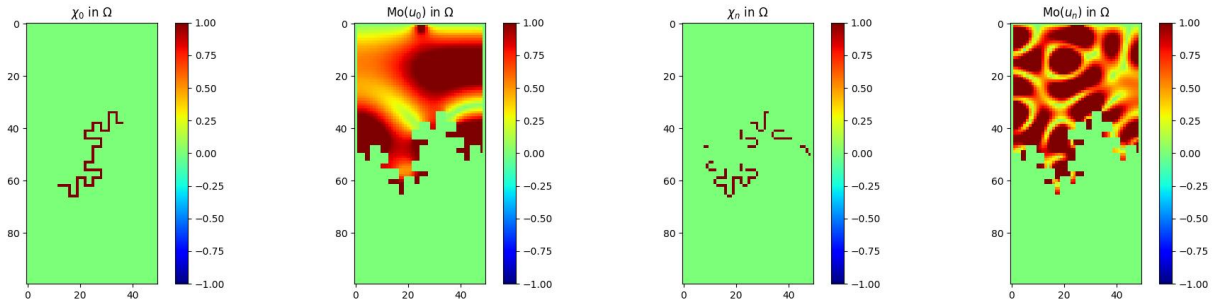
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Project	Academic team project, CentraleSupélec; team report of 7 November 2025 (original in French)
Team	T. Amiel, P. Blanc, C. Castillon, A. Chaibi, H. De Ruffray, S. El Mazlouzi
This document	The team project presented from the angle of my own contributions; the rest of the work is summarised for context
Contributions	Source model: T. Amiel. Mathematical analysis and open-air extension: T. Amiel with P. Blanc

Abstract

Treating a concert hall with absorbing material is expensive, and covering every wall is rarely possible. This project asks where a fixed quantity of porous material should be placed, on what wall shape and in what quantity, to minimise the acoustic energy in the hall. The hall is modelled by a Helmholtz boundary-value problem in which the material enters through a complex Robin condition weighted by its density χ on the wall. The energy $J(\chi) = \int_\Omega |p|^2$ is minimised by projected gradient descent under a mass constraint, the gradient $J'(\chi) = -\mathrm{Re}\{\alpha\, p\, q\}$ being obtained from an adjoint problem for the price of two PDE solves. Over 260 Hz–1.2 kHz, the team retained mineral wool covering 50 % of a level-2 pre-fractal wall; on a flat wall, mineral wool lowers resonance-peak energy by about two orders of magnitude relative to a solid panel. This report develops my contributions: a physically grounded source model (directivity, spectral lines, harmonics), and, with P. Blanc, the well-posedness of the problem with an absorbing wall, the adjoint gradient formula, and their extension to an open-air hall.

Keywords: Helmholtz equation; complex Robin condition; adjoint method; shape and material optimisation; fractal boundary; architectural acoustics.



(a) initial layout χ_0 (b) field $|u_0|$ (c) optimised layout χ_n (d) field $|u_n|$

Figure 1. Before (a, b) and after (c, d) optimisation, under the same mass constraint: the layout becomes fragmented and localised, and the field more dispersed with lower peak amplitudes.

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1 The project in context

1.1 Problem and state of the art

In existing buildings, acoustic correction is constrained by cost, aesthetics and heritage. Three families of solutions coexist. *Adding absorbing material* to existing walls is the most direct: covering 30–50 % of walls and ceilings is commonly recommended, for a reduction of 4–8 dB [1]; it needs no structural work but treats low frequencies poorly and does not guarantee a homogeneous field. *Designing the geometry* of the walls (diffusing or pseudo-fractal surfaces) disperses reflections; the Philharmonie de Paris [2] is the reference, with outstanding performance but heavy cost and lead time, so it is reserved for large new projects. *Decorative or heritage-compatible panels* (for instance perforated wood) reduce reverberation without altering the place.

The project combines the first two ideas at low cost: rather than covering the walls entirely with an expensive material, *optimise the spatial distribution of a fixed quantity of porous material, and the shape of the wall that carries it*, so as to minimise the acoustic energy in the whole hall. Room-in-room isolation, active cancellation and diffusers are outside this scope, since the goal is to minimise energy, not to homogenise the field.

1.2 Scope of this report and contributions

The project was carried out by a team of six; the original report is in French and covers the whole study. This document is organised around the parts I wrote, and summarises the others so that the work reads as a whole.

Component	Authorship	Section
State of the art, formulation, numerical method, parametric study	other team members (summary)	1
Source model: directivity, spectral lines, harmonics, second-chord check	**T. Amiel**	2
Well-posedness, lifting, adjoint gradient (closed hall)	**T. Amiel with P. Blanc**	3
Extension to an open-air hall	**T. Amiel with P. Blanc**	4

The theoretical study serves the numerical work in two ways: it fixes the functional framework in which the model is well posed, and it yields the adjoint gradient formula used by the optimiser.

1.3 Formulation

Let Ω be the (two-dimensional) hall, Γ_{abs} the wall that receives the material, $\chi : \Gamma_{\text{abs}} \to [0,1]$ the density of material ($\chi = \mathbf{1}_E$ for a binary layout) and $\beta \in [0,1]$ the proportion of wall treated. The pressure p_χ solves the

Helmholtz problem of Section 3. For a given material, frequency ω, wall shape and quantity β, the problem is

$$\text{find } \chi_{\text{opt}} \in U_{\text{ad}}(\beta) \;\text{ such that }\; J(\chi_{\text{opt}}) = \min_{\chi \in U_{\text{ad}}(\beta)} J(\chi), \qquad J(\chi) = \int_{\Omega} |p_\chi|^2 \,\mathrm{d}x,$$

with $U_{\text{ad}}(\beta) = \{\chi \in L^\infty(\Gamma_{\text{abs}}; [0,1]) : \frac{1}{|\Gamma_{\text{abs}}|}\int_{\Gamma_{\text{abs}}} \chi = \beta\}$ (a relaxed formulation; binarisation is done afterwards). A gradient descent on β itself is impractical, because of the way J depends on β and the numerical cost, which motivates the cascade below.

1.4 Numerical method (team)

- **Gradient.** Two Helmholtz solves per iteration (state p and adjoint q) give $J'(\chi) = -\operatorname{Re}\{\alpha\, p\, q\}$.
- **Projected descent under a mass constraint.** A step is accepted only if the energy decreases (step $\times 1.1$ on success, $\div 2$ on failure); the iterate is projected onto $\{0 \le \chi \le 1,\ \text{mean} = \beta\}$ by bisection on a uniform shift.
- **Boundary nodes.** On the grid, the gradient is evaluated at the interior nodes adjacent to the wall and transferred to the Robin nodes by neighbour averaging.
- **Binarisation.** Material is placed where χ is largest, without ever exceeding the available quantity.
- **Several frequencies.** The cost $J_{\text{total}} = \sum_k w_k \int_\Omega |u(\chi, k)|^2$ weights each critical frequency by its initial energy.
- **Material.** The complex absorption coefficient $\alpha(\omega)$ of each material is determined beforehand and held fixed during the topological optimisation.

1.5 Design cascade and results

The parameters are chosen one after the other, each time on top of the previous choices (Figure 2).

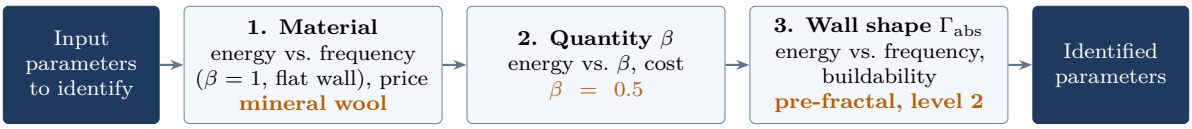


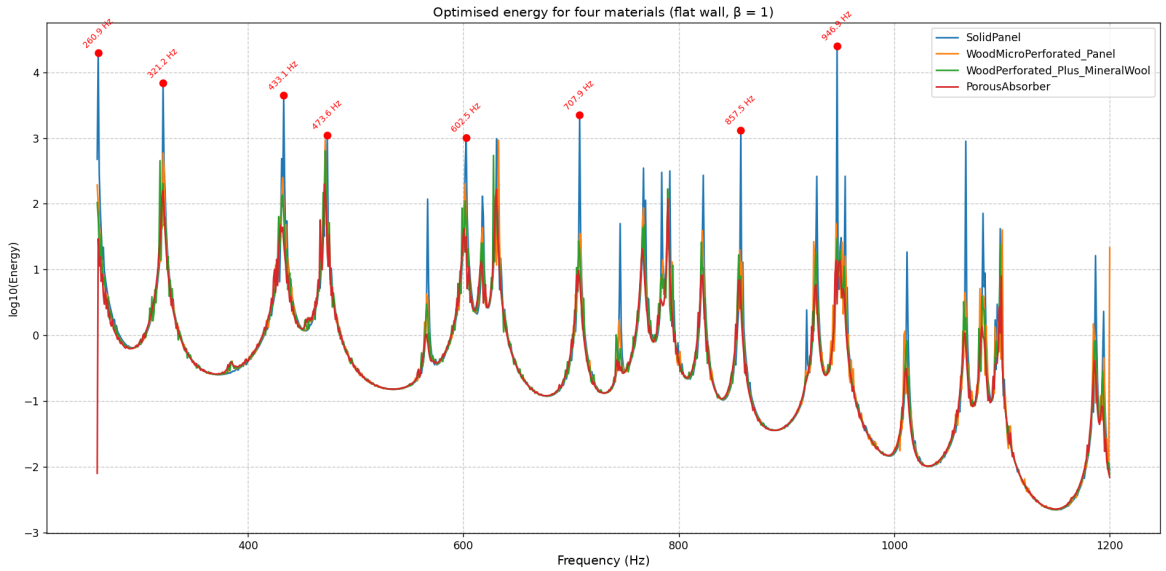
Figure 2. Cascade of design choices (the quantity β and the wall shape are optimised with the material already fixed).

Numbers behind the first two choices. The values below are *approximate*: they are read off the plots of the team report (Figure 3), not taken from tabulated data.

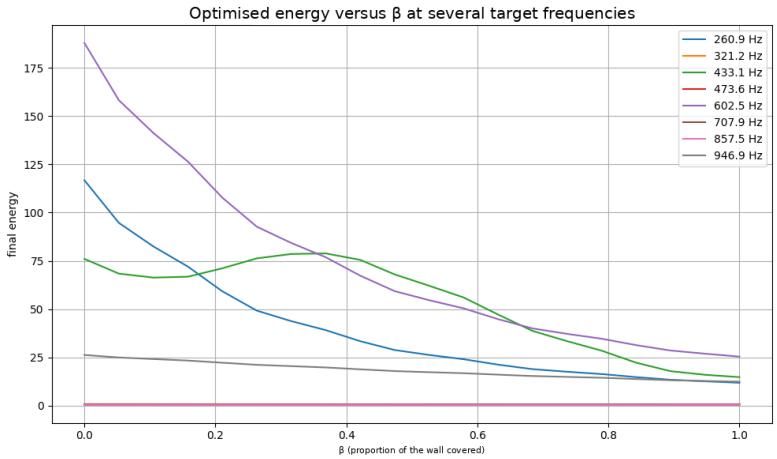
Energy J against β (arbitrary units)				Peak of $\log_{10}$(energy), flat wall		
Frequency	$\beta = 0$	$\beta = 0.5$	$\beta = 1$	Frequency	Solid	Mineral wool
260.9 Hz	117	28	12	260.9 Hz	4.3	1.5
433.1 Hz	76	65	15	433.1 Hz	3.65	1.7
602.5 Hz	188	58	26	602.5 Hz	3.0	1.6
946.9 Hz	27	18	12.5	946.9 Hz	4.4	1.15

Material choice: over the eight target frequencies the porous absorber (mineral wool) lowers the peak by 0.8 to 3.2 decades relative to the solid panel, a median of about two decades. Quantity: half of the wall already provides most of the benefit of full coverage at 260.9, 602.5 and 946.9 Hz (roughly 60–85 % of the reduction), whereas 433.1 Hz needs more material; the energy is not monotonic in β at that frequency up to $\beta \approx 0.4$.

For the wall shape, pre-fractal curves of levels 0 to 2 were compared; from level 3 on, the wall becomes difficult to build for an efficiency only slightly above that of level 2. The final design is therefore a level-2 pre-fractal wall half-covered with mineral wool (Figure 1).



(a) Four materials on a flat wall ($\beta = 1$): the porous absorber generally gives the lowest resonance peaks.



(b) Energy against the quantity β, at several target frequencies.

Figure 3. Evidence behind the first two choices (team results). Beyond $\beta = 0.5$, further reductions of energy are small compared with those obtained at lower β.

2 Source model

This section is my sole contribution.

2.1 Why the source matters, and how I built it

The optimised wall is computed from the energy that the source puts in the hall, so the result is only as realistic as the excitation. I worked iteratively. First came a deliberately minimal source, a pure isotropic note, enough to validate the optimisation loop quickly and check that the results were not aberrant. I then enriched it along three axes, each motivated by the physics of musical instruments.

- **Directivity.** Instruments radiate more directively as frequency rises (Özgencil and Yüğrük Akdağ [3]). This is modelled by a Gaussian in the position y along the stage, centred on the orchestra position y_0, whose width decreases like $1/\omega$.
- **Spectral spreading.** A played note is not a Dirac mass: the imprecision of the player and of the instrument spreads it slightly over neighbouring frequencies. Each note is therefore a narrow Gaussian line.
- **Harmonics.** Each instrument contributes harmonics, at prescribed relative levels.

2.2 The model

In the harmonic regime (time dependence $e^{-\mathrm{i}\omega t}$ understood), the hall is driven at frequency ω by the stage profile

$$g(y,\omega) = \sum_{i=1}^{9} A_i \exp\Big[-\frac{(\omega-\omega_i)^2}{\Delta\omega_i^2}\Big] \exp\Big[-\frac{(y-y_0)^2}{\ell_i(\omega)^2}\Big], \qquad \ell_i(\omega) = L_0\,\frac{\omega_i}{\omega+\omega_i}, \tag{1}$$

where $\omega_i = 2\pi f_i$ are the line frequencies of Table 1, A_i their amplitudes (from the relative levels of the table) and $\Delta\omega_i$ the width of line i. The spatial parameters are $y_0 = 14\,\mathrm{m}$ (middle of the stage), $L_0 = 0.8\,\mathrm{m}$, the characteristic size of an instrument (typically 0.5 to 1 m), and $c_0 = 343\,\mathrm{m\,s^{-1}}$, with $k = \omega/c_0$. The width $\ell_i(\omega)$ equals $L_0/2$ at the note's own frequency and decays like $1/\omega$ above it. The source has a single degree of freedom, the position along the stage; g is the Dirichlet datum on the stage Γ_{in} of the model of Section 3.

Frequency (Hz)	Note	Origin	Relative level (dB)
261.63	C_4	fundamental of C	0
329.63	E_4	fundamental of E	0
392.00	G_4	fundamental of G	0
523.25	C_5	harmonic 2 of C	−10
659.26	E_5	harmonic 2 of E	−10
784.00	G_5	harmonic 2 of G	−10
784.88	G_5	harmonic 3 of C	−15
988.89	B_5	harmonic 3 of E	−15
1176.00	D_6	harmonic 3 of G	−15

Table 1. Spectrum of the C-major chord: fundamentals and two harmonics each. Two lines nearly coincide near 784 Hz (the second harmonic of G and the third of C, 0.9 Hz apart).

2.3 Frequency band

Music spans 20 Hz to 20 kHz, far too wide for a first study, and low frequencies (20–250 Hz) are generally the hardest to attenuate. I chose a C-major chord, the most classical one, and limited each note to its first three harmonics. This gives the band 260 Hz–1.2 kHz of Figure 4, on which the efficiency of the design can be observed.

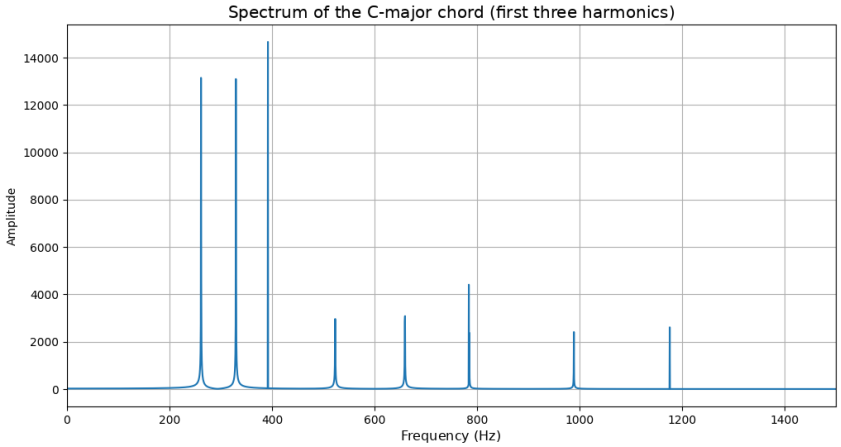
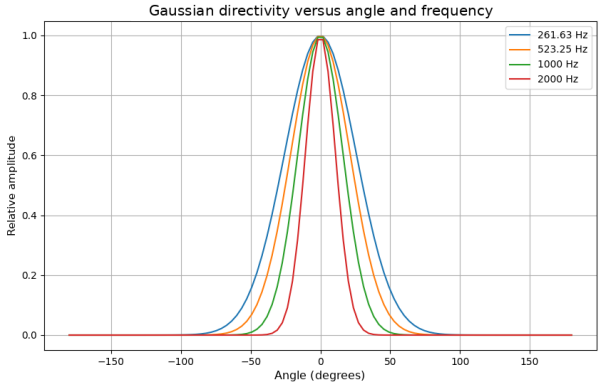


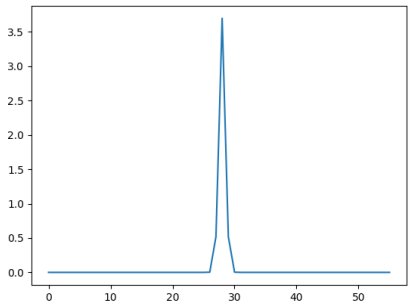
Figure 4. Spectrum of the source: C-major chord with its first three harmonics (Table 1).

2.4 Validation with a second chord

To check that the performance does not depend too much on the source, I repeated the computation with a C-minor chord (Figure 6). The curves for the two chords are similar, with the same positions of the energy peaks, which supports the view that the energy minimisation does not depend illegitimately on the source.



(a) The directivity sharpens as frequency increases.



(b) Resulting profile along the stage (grid-node index).

Figure 5. Gaussian directivity and spatial profile of the source.

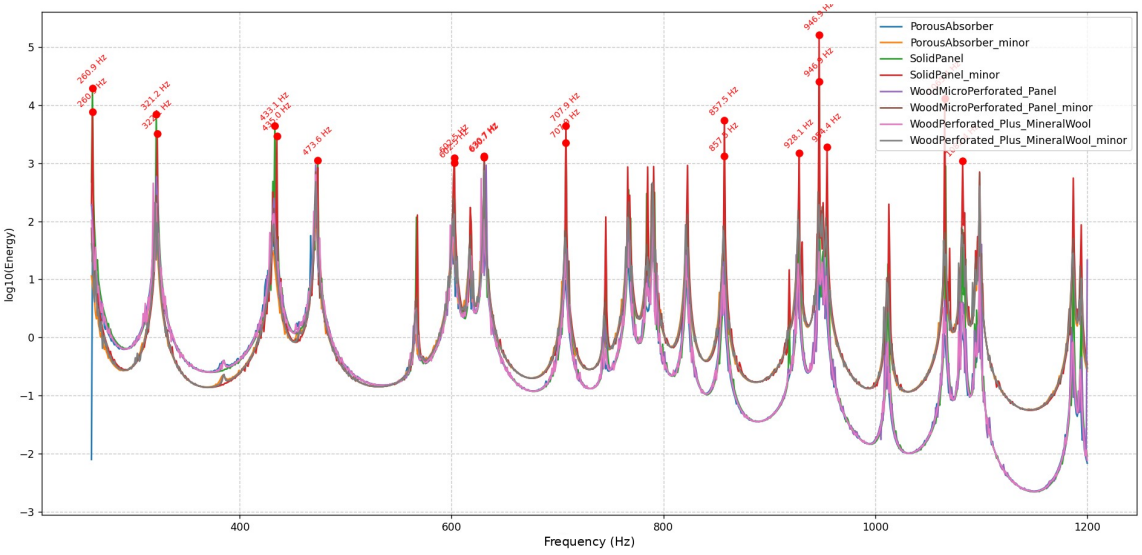


Figure 6. Energy against frequency for the final wall: each material appears for the C-major chord and for the C-minor chord (suffix `_minor`).

3 Mathematical model and analysis

Joint work with Paul Blanc.

3.1 Setting

The hall is a bounded domain $\Omega \subset \mathbb{R}^2$ (a $44\,\mathrm{m} \times 28\,\mathrm{m}$ slice of a shoebox hall modelled on the Concertgebouw), with $\partial\Omega = \Gamma_{\mathrm{in}} \cup \Gamma_{\mathrm{lat}} \cup \Gamma_{\mathrm{abs}}$ (Figure 7), outward normal n and $k = \omega/c_0 > 0$. The pressure p solves

$$\begin{cases} \Delta p + k^2 p = 0 & \text{in } \Omega, \\ \mathrm{Tr}(p) = g & \text{on } \Gamma_{\mathrm{in}}, \\ \partial_n p = 0 & \text{on } \Gamma_{\mathrm{lat}}, \\ \partial_n p + \alpha\chi\ \mathrm{Tr}(p) = 0 & \text{on } \Gamma_{\mathrm{abs}}. \end{cases} \tag{2}$$

The stage Γ_{in} emits the music through g (equation (1)); the side walls Γ_{lat} are rigid; on Γ_{abs} the porous material acts through a complex Robin condition with coefficient $\alpha \in \mathbb{C}$, weighted by the density $\chi \in L^\infty(\Gamma_{\mathrm{abs}}; [0,1])$.

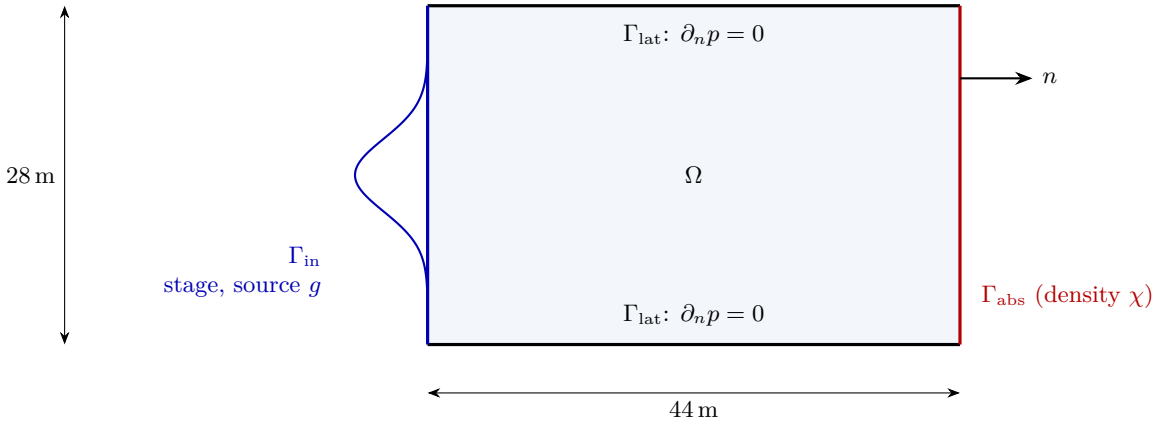


Figure 7. Slice of the hall at constant height (closed hall). The stage Γ_{in} emits; the wall Γ_{abs} receives the porous material; the side walls Γ_{lat} are rigid.

Functional framework. Let $V = \{v \in H^1(\Omega;\mathbb{C}) : \mathrm{Tr}_{\Gamma_{\mathrm{in}}} v = 0\}$, a Hilbert space for the H^1 norm (the preimage of the closed set $\{0\}$ under the continuous trace). The wall Γ_{abs} is a fractal curve, a d-set carrying the Hausdorff measure m_d; traces then take values in Besov spaces $B(\Gamma)$ of Jonsson–Wallin type [4], with dual $B'(\Gamma)$ and duality bracket $\langle\cdot,\cdot\rangle_\Gamma$. In the computations, Γ_{abs} is a level-2 pre-fractal, a polygonal curve for which Ω is Lipschitz and $B(\Gamma_{\mathrm{abs}}) = H^{1/2}(\Gamma_{\mathrm{abs}})$; the compactness of the trace $H^1(\Omega) \to L^2(\Gamma_{\mathrm{abs}})$ used below is stated in that setting. We assume that $g \in B(\Gamma_{\mathrm{in}})$ admits an extension $\tilde{g} \in B(\partial\Omega)$.

3.2 Lifting of the Dirichlet datum

To homogenise the condition on Γ_{in}, let $\hat{g}$ solve

$$\begin{cases} \Delta \hat{g} = 0 & \text{in } \Omega, \\ \mathrm{Tr}(\hat{g}) = g & \text{on } \Gamma_{\mathrm{in}}, \\ \partial_n \hat{g} = 0 & \text{on } \Gamma_{\mathrm{lat}} \cup \Gamma_{\mathrm{abs}}. \end{cases} \tag{3}$$

Proposition 1 (Lifting)**.** *Problem (3) has a unique weak solution $\hat{g} \in H^1(\Omega)$.*

Proof. A function with the right trace. By surjectivity of the trace there is $v \in H^1(\Omega)$ with $\mathrm{Tr}_{\partial\Omega} v = \tilde{g}$. Since $H^1(\Omega) = H^1_0(\Omega) \oplus V^1(\Omega)$ with $V^1(\Omega) = \{w \in H^1 : \Delta w = w\}$, write $v = v' + w$ with $v' \in H^1_0$: then $\Delta w = w$ and $\mathrm{Tr}_{\partial\Omega} w = \tilde{g}$, so $\mathrm{Tr}_{\Gamma_{\mathrm{in}}} w = g$.

Reduction to a coercive problem. Put $u = \hat{g} - w$. Then $-\Delta u = w$ in Ω, $\mathrm{Tr}_{\Gamma_{\mathrm{in}}} u = 0$ and $\partial_n u = -\partial_n w$ on $\Gamma_{\mathrm{lat}} \cup \Gamma_{\mathrm{abs}}$ (the normal derivative of w is well defined in the dual since $\Delta w = w \in L^2$). Green's formula, in which the term on Γ_{in} vanishes because $\mathrm{Tr}_{\Gamma_{\mathrm{in}}} \varphi = 0$, gives $a(u,\varphi) = \ell_w(\varphi)$ for all $\varphi \in V$, with

$$a(u,\varphi) = \int_\Omega \nabla u \cdot \overline{\nabla \varphi}, \qquad \ell_w(\varphi) = \int_\Omega w\,\overline{\varphi} - \big\langle \partial_n w, \overline{\mathrm{Tr}\,\varphi} \big\rangle_{\Gamma_{\mathrm{abs}} \cup \Gamma_{\mathrm{lat}}}.$$

Both forms are continuous. Since Γ_{in} has positive length, the Poincaré inequality holds on V, $\|u\|_{L^2} \leq C_P \|\nabla u\|_{L^2}$, hence $\operatorname{Re} a(u,u) = \|\nabla u\|_{L^2}^2 \geq (1+C_P^2)^{-1}\|u\|_{H^1}^2$: a is coercive on V. Lax–Milgram gives a unique u, hence $\hat{g} = u + w$. □

3.3 Well-posedness of the acoustic problem

Set $\hat{p} = p - \hat{g} \in V$. Since $\Delta \hat{g} = 0$ and $\partial_n \hat{g} = 0$ on Γ_{abs}, problem (2) becomes

$$\begin{cases} \Delta \hat{p} + k^2 \hat{p} = -k^2 \hat{g} & \text{in } \Omega, \\ \operatorname{Tr}(\hat{p}) = 0 & \text{on } \Gamma_{\text{in}}, \qquad \partial_n \hat{p} = 0 \ \text{ on } \Gamma_{\text{lat}}, \\ \partial_n \hat{p} + \alpha\chi \operatorname{Tr}(\hat{p}) = -\alpha\chi \operatorname{Tr}(\hat{g}) & \text{on } \Gamma_{\text{abs}}, \end{cases} \tag{4}$$

that is, $b_\chi(\hat{p}, \varphi) = L_\chi(\varphi)$ for all $\varphi \in V$, with

$$b_\chi(p,\varphi) = \int_\Omega \nabla p \cdot \overline{\nabla \varphi} - k^2 \int_\Omega p\,\overline{\varphi} + \alpha \int_{\Gamma_{\text{abs}}} \chi \operatorname{Tr} p \, \overline{\operatorname{Tr} \varphi}\, \mathrm{d}m_d,$$
$$L_\chi(\varphi) = k^2 \int_\Omega \hat{g}\,\overline{\varphi} - \alpha \int_{\Gamma_{\text{abs}}} \chi \operatorname{Tr} \hat{g}\, \overline{\operatorname{Tr} \varphi}\, \mathrm{d}m_d.$$

The term $-k^2 \int |p|^2$ destroys coercivity, so Lax–Milgram does not apply. The Fredholm alternative reduces the question to injectivity.

Proposition 2 (Well-posedness)**.** *Let Ω be Lipschitz (finite pre-fractal level) and assume*

(H1) $\operatorname{Im} \alpha \neq 0$*;*
(H2) $\chi > 0$ *almost everywhere on a non-empty relatively open set* $\Sigma \subset \Gamma_{\text{abs}}$*.*

Then for every $g \in B(\Gamma_{\text{in}})$, problem (4) has a unique solution $\hat{p} \in V$, so $p = \hat{p} + \hat{g}$ is the unique solution of (2) in $H^1(\Omega)$, and $\|p\|_{H^1} \leq C \|g\|_{B(\Gamma_{\text{in}})}$ with $C = C(k, \alpha, \chi, \Omega)$.

Proof. Fredholm structure. Let $\mathfrak{B}, \mathfrak{B}_0 : V \to V'$ be the operators of b_χ and of $\int \nabla p \cdot \overline{\nabla \varphi}$. $\mathfrak{B}_0$ is invertible (Lax–Milgram, Poincaré) and $\mathfrak{B} - \mathfrak{B}_0$ is compact: the embedding $H^1(\Omega) \hookrightarrow L^2(\Omega)$ and the trace $H^1(\Omega) \to L^2(\Gamma_{\text{abs}})$ are compact for Lipschitz Ω, and $0 \leq \chi \leq 1$. By the Fredholm alternative [5], $\mathfrak{B}$ is invertible if and only if it is injective.

Injectivity. Let $p \in V$ with $b_\chi(p, \varphi) = 0$ for all $\varphi \in V$. Taking $\varphi = p$:

$$\int_\Omega |\nabla p|^2 - k^2 \int_\Omega |p|^2 + \alpha \int_{\Gamma_{\text{abs}}} \chi |\operatorname{Tr} p|^2 = 0.$$

The imaginary part gives $\operatorname{Im} \alpha \int_{\Gamma_{\text{abs}}} \chi |\operatorname{Tr} p|^2 = 0$, so by (H1) $\chi \operatorname{Tr} p = 0$ almost everywhere on Γ_{abs}, and by (H2) $\operatorname{Tr} p = 0$ on Σ. The Robin condition then gives $\partial_n p = -\alpha\chi \operatorname{Tr} p = 0$ on Σ: the Cauchy data of p vanish on Σ. Extend p by 0 across Σ into a small ball B with $B \cap \partial\Omega \subset \Sigma$. The extension is a weak solution of $\Delta p + k^2 p = 0$ in the connected open set $\Omega \cup B$ and vanishes on the non-empty open set $B \setminus \overline{\Omega}$. Solutions of the Helmholtz equation are real-analytic, so $p \equiv 0$.

Continuous dependence follows from the invertibility of $\mathfrak{B}$ and the continuity of $g \mapsto \hat{g}$. □

Remark (Resonances and sign convention). The constant C is not uniform. As $\chi \to 0$ the problem tends to a cavity with rigid walls, whose uniqueness fails at the eigenfrequencies of the mixed Laplacian (Dirichlet on Γ_{in}, Neumann elsewhere), and C can become large nearby: this is consistent with the energy peaks observed against frequency (Figure 6). Hypothesis (H1) says that the material is strictly dissipative: with the convention $e^{-\mathrm{i}\omega t}$, the outward energy flux through Γ_{abs} is proportional to $-\operatorname{Im} \alpha \int \chi |p|^2$, so the material absorbs when $\operatorname{Im} \alpha < 0$.

3.4 Gradient of the energy by the adjoint method

Let $B_\chi(u,\varphi) = \int_\Omega \nabla u \cdot \nabla \varphi - k^2 \int_\Omega u\varphi + \alpha \int_{\Gamma_{\text{abs}}} \chi\, u\varphi \, \mathrm{d}m_d$ be the *bilinear* (no conjugation) form associated with the same operator; it is symmetric, $B_\chi(u,\varphi) = B_\chi(\varphi,u)$.

Proposition 3 (Gradient)**.** *Under (H1)–(H2), $\chi \mapsto p_\chi$ is Fréchet differentiable and $J(\chi) = \int_\Omega |p_\chi|^2$ satisfies, for every admissible perturbation $\delta\chi$,*

$$\boxed{\delta J = -\operatorname{Re} \int_{\Gamma_{\text{abs}}} \alpha\, p_\chi\, q_\chi\, \delta\chi \, \mathrm{d}m_d, \qquad \textit{i.e. } J'(\chi) = -\operatorname{Re}\{\alpha\, p_\chi\, q_\chi\} \ \textit{ on } \Gamma_{\text{abs}},} \tag{5}$$

where the adjoint state $q_\chi \in V$ solves

$$\begin{cases} \Delta q + k^2 q = -2\,\overline{p}_\chi & \text{in } \Omega, \\ \operatorname{Tr}(q) = 0 & \text{on } \Gamma_{\text{in}}, \qquad \partial_n q = 0 \quad \text{on } \Gamma_{\text{lat}}, \\ \partial_n q + \alpha\chi \operatorname{Tr}(q) = 0 & \text{on } \Gamma_{\text{abs}}. \end{cases} \tag{6}$$

Proof. Differentiability. $\chi \mapsto \mathfrak{B}_\chi \in \mathcal{L}(V, V')$ and $\chi \mapsto L_\chi \in V'$ are affine and continuous on $L^\infty(\Gamma_{\text{abs}})$, and inversion is smooth on invertible operators. By Proposition 2, $\hat{p}_\chi = \mathfrak{B}_\chi^{-1} L_\chi$ is therefore Fréchet differentiable, and so is $p_\chi = \hat{p}_\chi + \hat{g}$ since $\hat{g}$ does not depend on χ.

Sensitivity. Differentiating $B_\chi(\hat{p}, \varphi) = k^2 \int \hat{g}\varphi - \alpha \int_{\Gamma_{\text{abs}}} \chi\, \hat{g}\varphi$ in the direction $\delta\chi$ and using $p = \hat{p} + \hat{g}$ gives, for all $\varphi \in V$,

$$B_\chi(\delta p, \varphi) = -\alpha \int_{\Gamma_{\text{abs}}} \delta\chi\, p\, \varphi \, \mathrm{d}m_d.$$

(In strong form: $\Delta\delta p + k^2 \delta p = 0$, with $\partial_n \delta p + \alpha\chi\, \delta p = -\alpha\, \delta\chi\, p$ on Γ_{abs}.)

Adjoint. The weak form of (6) is $B_\chi(q, \varphi) = 2\int_\Omega \overline{p}\,\varphi$ for all $\varphi \in V$; q exists and is unique because $\mathfrak{B}_\chi$ is invertible and symmetric. Taking $\varphi = \delta p$ in the adjoint equation and $\varphi = q$ in the sensitivity equation, and using the symmetry of B_χ,

$$2\int_\Omega \overline{p}\,\delta p = B_\chi(q, \delta p) = B_\chi(\delta p, q) = -\alpha \int_{\Gamma_{\text{abs}}} \delta\chi\, p\, q \, \mathrm{d}m_d.$$

Finally $\delta J = 2\operatorname{Re}\int_\Omega \overline{p}\,\delta p = -\operatorname{Re}\int_{\Gamma_{\text{abs}}} \alpha\, \delta\chi\, p\, q \, \mathrm{d}m_d$. □

Two remarks. First, because B_χ is symmetric rather than Hermitian, the adjoint operator coincides with the direct one and only the source changes, to the conjugate $-2\overline{p}$ (using $-2p$ would give a wrong gradient). Second, the gradient with respect to *all* points of the wall costs two Helmholtz solves, which is what makes the descent affordable.

4 Extension to an open-air hall

Joint work with Paul Blanc.

The side walls are now open: energy can leave the domain. We model the opening by the first-order absorbing radiation condition $\partial_n p = \mathrm{i}k \operatorname{Tr} p$ on Γ_{lat} (a simplified form of the Sommerfeld condition, exact for normal incidence; with the convention $e^{-\mathrm{i}\omega t}$ it selects outgoing waves). The direct problem is

$$\begin{cases} \Delta p + k^2 p = 0 & \text{in } \Omega, \\ \operatorname{Tr}(p) = g & \text{on } \Gamma_{\text{in}}, \\ \partial_n p = \mathrm{i}k \operatorname{Tr}(p) & \text{on } \Gamma_{\text{lat}}, \\ \partial_n p + \alpha\chi \operatorname{Tr}(p) = 0 & \text{on } \Gamma_{\text{abs}}. \end{cases} \tag{7}$$

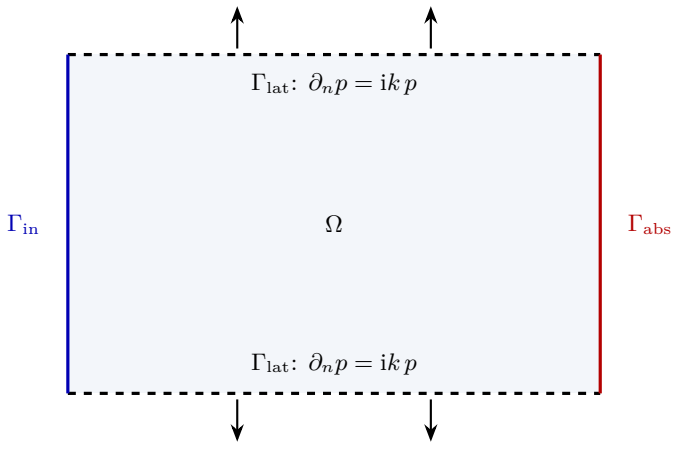


Figure 8. Open-air configuration: the side walls let the energy escape (arrows).

Proposition 4 (Open-air hall)**.** *Assume* $\operatorname{Im}\alpha \le 0$ *(a dissipative or neutral material). Then* (7) *has a unique solution in* $H^1(\Omega)$ *for every* $g \in B(\Gamma_{\text{in}})$*, and the gradient of* J *is given by the same formula* (5)*, where* p_χ *and* q_χ *now solve* (7) *and its adjoint, which differs from* (6) *only by the condition* $\partial_n q = \mathrm{i}k \operatorname{Tr} q$ *on* Γ_{lat}*.*

Proof. Lifting. The lifting satisfies $\Delta\hat{g} = 0$, $\operatorname{Tr}\hat{g} = g$ on Γ_{in}, $\partial_n\hat{g} = \mathrm{i}k \operatorname{Tr}\hat{g}$ on Γ_{lat} and $\partial_n\hat{g} = 0$ on Γ_{abs}. With the function w of Proposition 1 and $u = \hat{g} - w$, one has $-\Delta u = w$, $\operatorname{Tr}_{\Gamma_{\text{in}}} u = 0$, $\partial_n u = -\partial_n w$ on Γ_{abs} and $\partial_n u = \mathrm{i}k(\operatorname{Tr} u + \operatorname{Tr} w) - \partial_n w$ on Γ_{lat}. Green's formula leads to $a(u,\varphi) = \ell_w(\varphi)$ on V with

$$a(u,\varphi) = \int_\Omega \nabla u \cdot \overline{\nabla\varphi} - \mathrm{i}k \int_{\Gamma_{\text{lat}}} \operatorname{Tr} u\, \overline{\operatorname{Tr}\varphi},$$
$$\ell_w(\varphi) = \int_\Omega w\,\overline{\varphi} + \mathrm{i}k \int_{\Gamma_{\text{lat}}} \operatorname{Tr} w\, \overline{\operatorname{Tr}\varphi} - \left\langle \partial_n w, \overline{\operatorname{Tr}\varphi} \right\rangle_{\Gamma_{\text{abs}}\cup\Gamma_{\text{lat}}}.$$

The added terms are continuous, and $-\mathrm{i}k\int_{\Gamma_{\text{lat}}} |\operatorname{Tr} u|^2$ is purely imaginary, so $\operatorname{Re} a(u,u) = \|\nabla u\|_{L^2}^2$: a is coercive and Lax–Milgram applies.

Well-posedness. The homogenised problem for $\hat{p} = p - \hat{g}$ has the same structure, and the additional term $-\mathrm{i}k\int_{\Gamma_{\text{lat}}} \operatorname{Tr} p\, \overline{\operatorname{Tr}\varphi}$ is compact, so the Fredholm argument is unchanged. For injectivity, the imaginary part of the energy identity reads

$$-k \int_{\Gamma_{\text{lat}}} |\operatorname{Tr} p|^2 + \operatorname{Im}\alpha \int_{\Gamma_{\text{abs}}} \chi |\operatorname{Tr} p|^2 = 0.$$

If $\operatorname{Im}\alpha \le 0$ both terms are non-positive, hence both vanish: $\operatorname{Tr} p = 0$ on Γ_{lat} and $\partial_n p = \mathrm{i}k \operatorname{Tr} p = 0$ there. The Cauchy data vanish on Γ_{lat}, and unique continuation gives $p \equiv 0$. Uniqueness thus follows from the radiation condition alone, with no assumption such as (H2).

Gradient. The bilinear form acquires the term $-\mathrm{i}k\int_{\Gamma_{\text{lat}}} u\varphi$, which is still symmetric and does not depend on χ. The proof of Proposition 3 therefore goes through verbatim. In Green's identity, the contribution of Γ_{lat} cancels: $\partial_n q\,\delta p - q\,\partial_n\delta p = \mathrm{i}k\,q\,\delta p - \mathrm{i}k\,q\,\delta p = 0$. □

Interpretation. In both cases the gradient has the same form; only the conditions on Γ_{lat} change, and they model different physics: walls for the closed hall, and wave leakage for an open hall. The open-air case would apply to an outdoor concert venue or to the placement of an absorbing liner in an aircraft engine duct. The remainder of the project stays with reflecting side walls, the case of a traditional concert hall.

5 Conclusion and extensions

The study establishes the functional framework of the model and the adjoint gradient formula that motivate the algorithm. The source model gives the optimisation a physically grounded excitation, whose robustness was checked on a second chord, and the same gradient formula covers the closed hall and the open-air configuration. Several natural extensions follow.

- **Continuous and discrete adjoints.** Derive the adjoint of the discretised system on the grid and validate the gradient by finite differences, in particular at the Robin boundary nodes, to tie the continuous formula to the implementation.
- **From pre-fractal to fractal.** Establish constants (Poincaré, trace, stability) that are uniform in the pre-fractal level, to connect the level-2 computations with the d-set framework.
- **Existence of minimisers.** Weak-$*$ compactness of $U_{\text{ad}}(\beta)$ together with the continuity of $\chi \mapsto p_\chi$ (compact trace) is the natural route to existence on the relaxed set; the effect of binarisation on the value of J is a companion question.
- **Near resonances.** Quantify how the stability constant depends on k and χ, and what this implies for the multi-frequency optimisation.
- **Absorption coefficient.** Relate $\alpha(\omega)$ to the effective properties of the porous medium and to the dissipation condition $\operatorname{Im}\alpha \neq 0$.
- **Realism and applications.** Richer sources (measured directivities, several instruments), three dimensions, and the open-air applications of Section 4.

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